Experiment 4 Spectrum Analysis of Various Signals

I. Experimental Objectives

(1) Use a spectrum analyzer to observe various actual signals.

(2) Use the fast Fourier transformation (FFT) to observe the frequency response.

(3) Understand the relationship between the time-domain and frequency-domain characteristics of various signals.

II. Requirements for Pre-experiment Preparation

(1)What are the formulas to convert between linear and logarithmic coordinates? What is the gain when the voltage gain is -6 dB? What is the attenuation when the power attenuation is 3 dB?

(2) Multiply a sine wave by a square wave to generate a half-wave rectified sine wave and calculate its spectrum.

III. Experiment Principle

When performing time-domain to frequency-domain transformation on discrete signals, the Fast Fourier Transform (FFT) is commonly used. FFT is an improved algorithm of the Discrete Fourier Transform based on its odd, even, imaginary, and real characteristics. It still relies on the Fourier Transform but simplifies the complex polynomial changes of the Discrete Fourier Transform by using a series of mathematical algorithms. Its function remains to convert signals from the time domain into the frequency domain. The frequency spectrum in this experiment are all calculated by FFT based on the time-domain waveforms observed on the oscilloscope.

IV. Experimental Procedures

1.Spectrum of pulse sequence

Connect the circuit as shown in Figure 5.1 and observe the time-domain and frequency-domain waveforms of the pulse sequence through Lab 9 tab. Record them in Figure 5.2. The settings of each parameter are as follows:

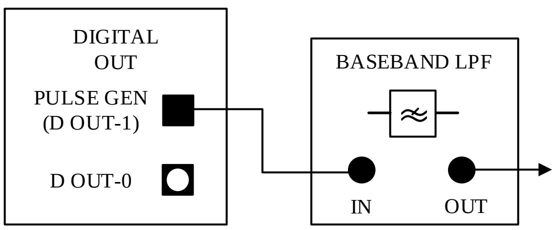

Figure 5.1 Pulse sequence after baseband filtering

①Pulse Generator (PULSE/CLK GENERATOR): Frequency is 500 Hz; Duty cycle is 10%.

②Connect CH1 of the oscilloscope to the system input and CH0 of the oscilloscope to the output of the baseband filter.

Set the Y-axis scale (GAIN) of the FFT-X9 on the right-hand side to linear mapping (right-click, 显示项(display item)-标尺图列(cursor legend), then the icons related to FREQUENCY and GAIN appear on the side, and then click the last identifier on the right, and select the mapping mode - linear).

The spectrum analyzer has a linear frequency axis and a linear scale on the vertical axis. When the time base (TIMEBASE) of the oscilloscope changes, the frequency scale of the spectrum analyzer also changes accordingly. In subsequent experiments, it is necessary to adjust the time base to balance the resolution and display range of the oscilloscope and the spectrum analyzer.

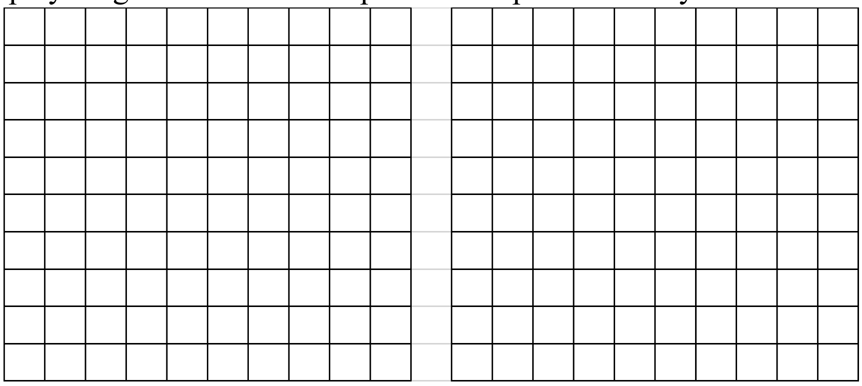

Figure 5.2 Time-domain and frequency-domain waveforms of the pulse sequence

Observe the pulse sequence, with a width of _____ms and a repetition period of ________ms.

Question 5.1

At which frequency is the gain of the envelope of the pulse sequence zero? (   )

A. 2 kHz.
B. 3 kHz.
C. 4 kHz.
D. 5 KHz.

By changing the duty cycle of the pulse sequence (e.g.20%), the pulse width varies. Under the condition of maintaining a constant frequency, it can be observed that the frequency interval between zero gain points changes with the variation of the pulse width.

Question 5.2

It is non-reciprocal relationship between the frequency interval between zero-gain points of the pulse sequence and the pulse width?

A. Yes.
B. No.

Question 5.3

When the duty cycle approaches 0, each frequency component in frequency spectrum has approximately the same gain, equivalent to the input pulse sequence being an impulse function?

C. Yes.
D. No.

2.Spectrum of the filtered pulse sequence

The settings for restoring the pulse generator are as follows:

① Pulse generator: frequency 500 Hz; duty cycle 10%.

② Connect CH1 of the oscilloscope to the input of the BLPF module and CH2 of the oscilloscope to the output of the BLPF module.

Observe the time-domain and frequency-domain waveforms of the impulse response at the output of the BLPF channel. Set the Y-scale mapping of the FFT display to logarithmic and record it in Figure 5.3.

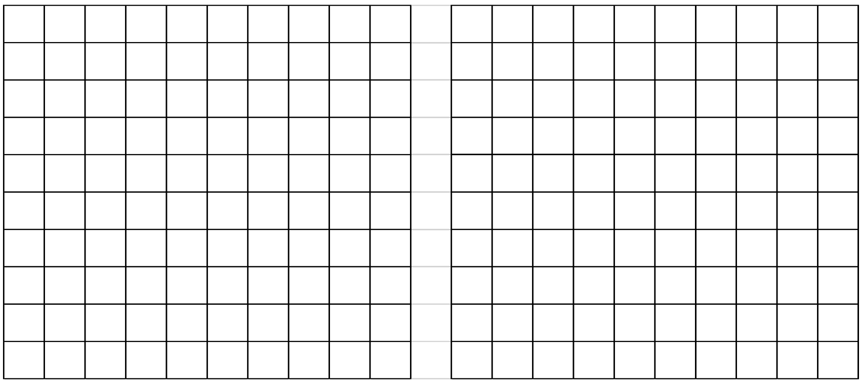

Figure 5.3 Time-domain and frequency-domain waveforms of BLPF output

Plot the response characteristic curves of the BLPF, with the vertical coordinate being expressed in logarithmic form (such as lg a), and the unit is dB.

3. "Sinc Pulse" sequence

In the previous experiment "Spectrum of Pulse Sequences", the spectrum of a rectangular pulse sequence is observed, and its characteristic is known as the "Sinc function". In the time domain, a rectangular signal generates a frequency response spectrum in the shape of a Sinc function. To verify the spectrum of the Sinc pulse, the DAC-1 module of the experimental board ANALOG OUT simulates a Sinc pulse sequence signal that is infinite in time domain but finite in frequency domain. However, in reality, a single Sinc pulse cannot be infinitely long in time but can only be approximately infinite. In this experiment, a repetitive pulse signal that terminates oscillation after approximately 20 cycles is generated for approximation. The characteristics of this signal are observed. Please plot the time and the frequency response of Sinc pulse signals in Figure 5.4.

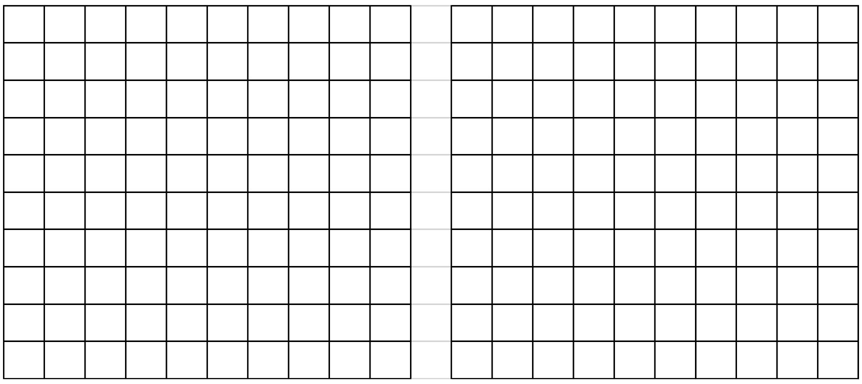

Figure 5.4 Time-domain and frequency-domain waveforms of the Sinc pulse

4. Nonlinear processes: Clipping, rectification and harmonic doubling

Due to overloading and other reasons, sinusoidal clipping or limiting often occurs in actual systems. Sometimes, maintaining the amplitude of the signal is not important (such as FM signals, etc.), because the amplitude of the signal is determined by the frequency and phase and does not carry other information.

Connect the circuit as shown in Figure 5.5 and set as follows:

① Function generator: Select sine wave output, with a frequency of 1 kHz and an amplitude of 4 Vpp.

② Oscilloscope: time base:10 ms.

Set the limiters' toggle switches to all possible positions and plot the observed results in Figure 5.6.

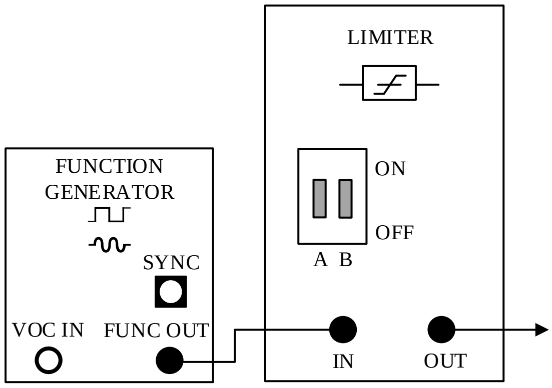

Figure 5.5 Wiring diagram of clipping of sine wave signal

Question 5.4

Limiting produces harmonics in spectrum of signals? (   )

A. Yes.

B. No.

V. Notes to Observe

(1) The basic precautions are the same as in Experiment 2.

(2) The time-domain and frequency-domain data of this experiment can be exported as pictures from the experimental program or exported as data and then plotted using Matlab.

VI. Requirements for the Experimental Report

(1) Complete the experimental content independently and record the experimental results honestly.

(2) Experimental reflection questions should be written in the experiment report, and experimental feelings, opinions and suggestions should be written after the experimental conclusion.